SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I M.Pavankalyan(192121478) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

level-3

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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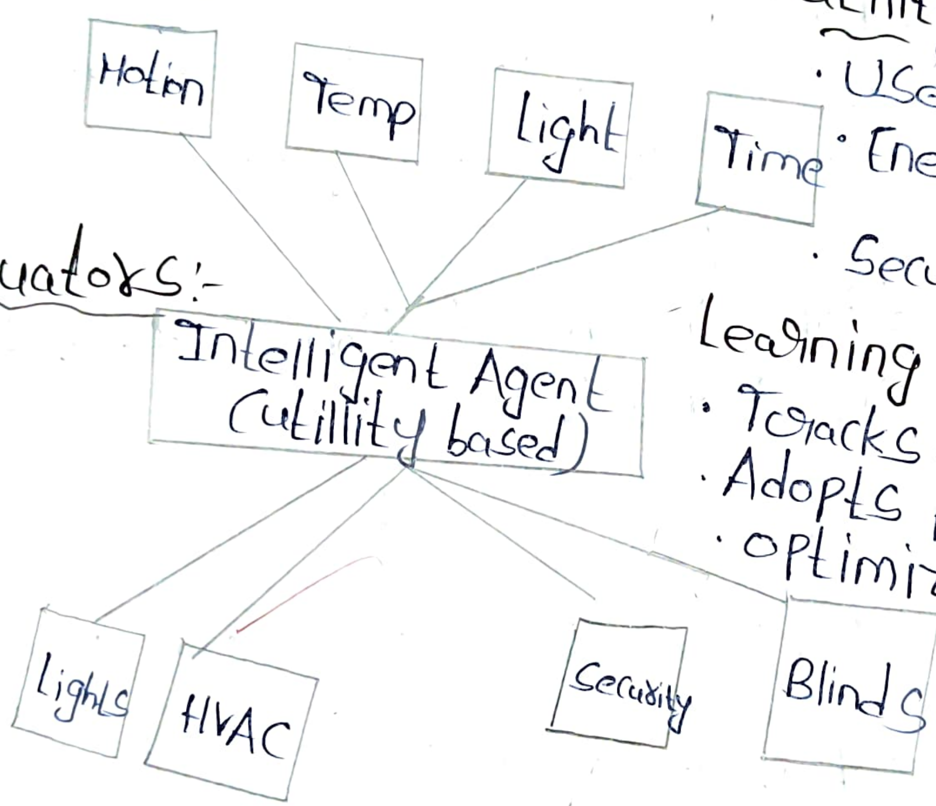
Descriptive Test Sub:- A.I

Smart Home Automation System:

A System home automation system is being designed to manage lighting temperature & Security based on user behaviour and environmental Conditions. The system must respond to real time i/ps such as motion detection, temperature changes and time of day while also learning user preferences overtime.

Architecture:-

Sensors:-



Actuators:-

utility calculations:
- User Comfort (30%)
- Energy η (40%)
- Security level (30%)

Learning Loop:
- Tracks user behaviour
- Adopts preferences
- optimizes decisions

Types of Intelligent Agents:-

Simple Reflex:-

- Condition action rules
- No memory
- Fast response

Model-Based:

- Maintains world model
- Predicts outcomes.
- Considers history

Goal Based:

- Has explicts goal
- Plans actions
- Searches for solutions.

Utility Based:-

- Maximizes utility.
- Evaluate trade offs.
- optimizes Performance.

Application in Smart Home Context:-

- Practical Scenario:- Consider a typical morning situation. The system detects motion in the bed room at 6:30 AM, temperature is 18°C and outdoor is low.
- Simple Reflex agent:- Instantly turns on lights and heating based only

n current sensor data.
Model based agent:- Plans actions to achieve comfort while reducing energy consumption.
Utillity Based agent:- Chooses the action with the highest overall utility by balancing comfort, n Security.
Recommended Approach:- Hybrid utility Based agent with learning.
· Comfort:- Learns user preferences and optimizes temperature and the lighting for maximum comfort.
· Energy n:- Reduces energy use by Scheduling Smart heating energy periods.
Security:- Continuously monitors security and gives higher priority to threats while maintaining comfort 4n.
Hybrid Architecture benefits:
· Model based core:- Maintains, historical data and predicts future states.

Goal based Component :- Handles Security & Critical Systems
Utility based :- optimizes routine operations
Learning mechanism :- Continuously adopts to charging user Perference and adaptability (learns optimal St settings over time)
Indtroduce Deegening Search :- IDS
Performs repeated depth first searches with increasing depth limits. It starts with depth limit 1, then 2, then 3, etc, omit the goals. This Combines memory of n of DFS with optimaliry of BPS.
How IDS works for Maze Navigation :
Set depth limit = 1
· Perform depth first search up to depth d.
· If goal, not found, increment depend repeat.
· Path length depends minimum moves to reach eat.

Time Complexity ($O(b^d)$ where d is solution depth Space Complexity: $O(d)$ (only stars nodes on current path)

Advantages & Disadvantages:-

Criterion	UCS	IDS
Optimality	optimal (finds min pat)	optimal.
Completeness	Complete (cost>0)	Complete
Time Complexity	$O(b^\wedge(C^*/\varepsilon)$	$O(b^\wedge d)$
Space Complexity	$O(b^\wedge(C^*/\varepsilon)$	$O(d)$: Low

2) Robot Navigation in Unknown Maze:

A Robot is placed in an unknown Maze & must find a path to the exist without prior knowledge of the layout. The robot must determine it's path using Search algorithms.

Uniformed Search Strategies Explanation

3) Uniform Cost Search (UCS): UCS expands nodes in order of their path cost from the root node, always exploring the the node problem. In the maze problem if cells have different transversal costs UCS finding the minimum cost path.

How UCS works for Maze Navigation:-

1. Start at entry point with cost=0
2. Maintain a priority queue sorted by Pat cost.
3. Expand lowest cost node, add neighbours to queue.
4. Path cost represents total movement cost to reach exit.

Time complexity: $O(b^{C^*/\varepsilon})$ wher C^* is optimal cost & ε is minimum edge cost.

Space complexity: $O(b^{C^*/\varepsilon})$ (stores all modes inqueue)

Implementation Strategy:-

Use IDS as primary search with depth limit increasing.

· Maintain a visited set to avoid.

re-exploring cells.
- If robot returns to know region, use model to shortcut navigation.
- Continue until exit is found.

Comparison with Alterative Approaches:-
- UCS:- Would waste memory storing all nodes with costs

Simple Reflex:- No planning; would require manual intervention ends.

Conclusion:- The recommended goal based agent using IDS provides an optimal balance of n memory usage; solution quality for maze navigation in unknown environments.